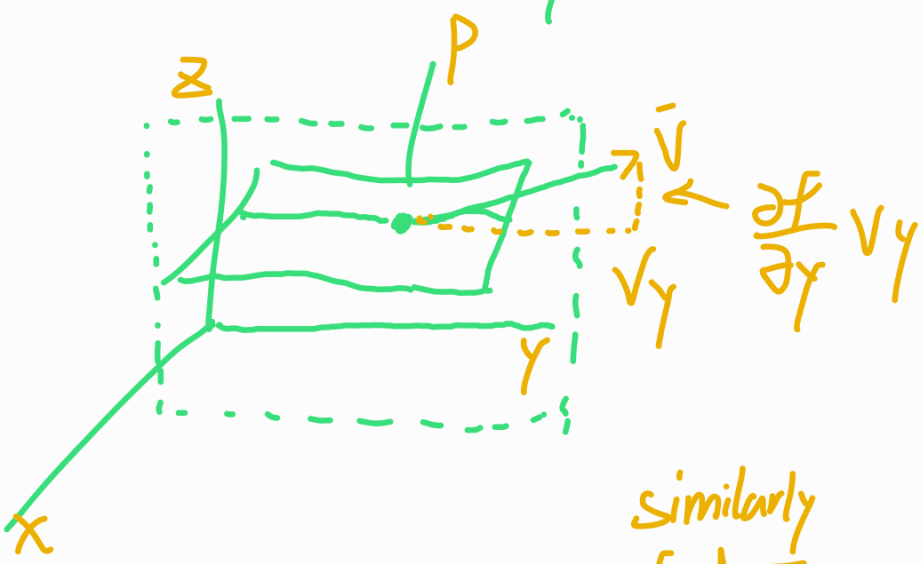
① Compute normal $\hat{n}$

Given $z = f(x,y)$

$$\hat{n} = \frac{u \times v}{|u \times v|} = \frac{-i\frac{\partial f}{\partial x} - j\frac{\partial f}{\partial y} + k}{\sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 + 1}}$$

note: down by



similarly find $\bar{u}$

② Surface Integral of a scalar fun

$$\iint G(x,y,z)\, dS$$

For Cartesian / Polar

$$\iint G[x,y,f(x,y)] \cdot \sqrt{1+(\partial f/\partial x)^2 + (\partial f/\partial y)^2}\, dx + dy$$

③ Dot Product Form

$$\iint F \cdot \hat{n}\, dS$$

$$= \iint \left[-F_x(\cdot)\frac{\partial f}{\partial x} - F_y(\cdot)\frac{\partial f}{\partial x} + F_z(\cdot) \right] dx\, dy$$

④ Divergence

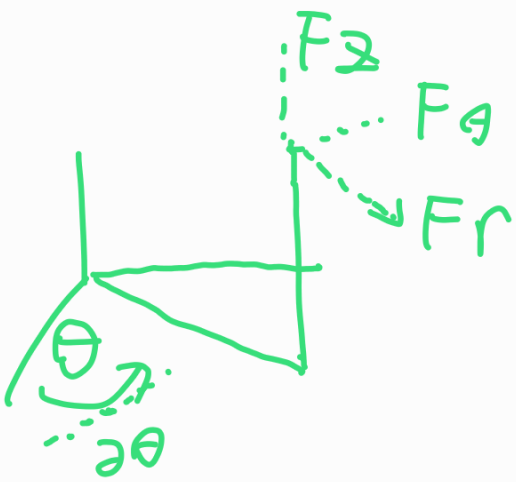
Compute by adding integral faces of a ΔV

Cartesian

$$\operatorname{div} F = i\frac{\partial F_x}{\partial x} + j\frac{\partial F_y}{\partial y} + k\frac{\partial F_z}{\partial z}$$

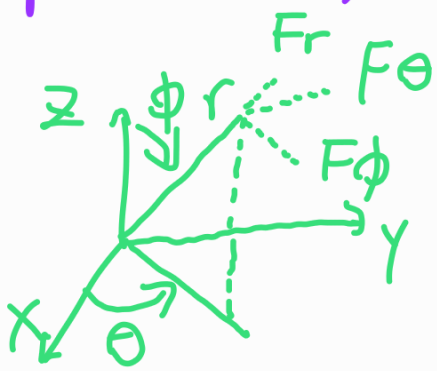
Cylindrical

$$\operatorname{div} F = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z}$$



Spherical

$$\operatorname{div} F = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 F_r\right) + \frac{1}{r\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi F_\phi\right) + \frac{1}{r\sin\phi}\frac{\partial F_\theta}{\partial \theta}$$



Del $\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

$$\nabla \cdot F = i\frac{\partial}{\partial x} \cdot F_x + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$$

$$= \operatorname{div} F$$

5 Divergence Theorem

$$\iint_S F \cdot \hat{n}\, dS = \iiint_V \nabla \cdot F\, dV$$